Object Detection with TensorFlow.js

Overview

This workshop implements an AI-powered object detection system integrated into Node-RED. The system allows users to upload images and automatically detects objects using a pre-trained machine learning model.

The goal is to demonstrate how artificial intelligence can be embedded into IoT workflows for real-time image analysis.

System Description

The system uses TensorFlow.js with the COCO-SSD pre-trained model to analyze images uploaded by the user. Once an image is submitted, the system processes it and returns detected objects with bounding boxes, labels, and confidence scores.

The results are displayed visually in the Node-RED dashboard along with a structured list of detected objects.

Implementation Details

The workflow is composed of several stages:

- A dashboard interface provides an image upload form
- The uploaded image is passed into the Node-RED flow
- A TensorFlow.js processing node loads the COCO-SSD model
- Object detection is performed on the image
- The system outputs:
    - Bounding boxes drawn on the image
    - Labels identifying detected objects
    - Confidence scores for each detection
- Results are displayed in both visual and textual formats in the dashboard

The processing is handled using Node.js integration with TensorFlow.js, ensuring compatibility within Node-RED, which demonstrates how AI-based computer vision can be integrated into IoT workflows, enabling intelligent data processing directly within Node-RED.